

"I see a reliability issue here ..."

- external API / LLM call
 - timeout + retry + backoff + exponential fallback
- response.json()["key"]
 - check status + validate before parsing
- missing error handling
 - try/except + proper HTTP codes

"I see a security concern here ..."

- pickle.loads(...)
 - unsafe deserialization / RCE
- f"...(user_input)..." SQL
 - SQL injection
- hardcoded API keys
 - env variables / secret manager
- LLM output execution
 - Pydantic validation

"I would split this into separate components..."

- God class/function
 - SRP violation

"This will be hard to test ..."

- direct HTTP/DB/filesystem
 - mocks + dependency injection
- service instantiated inside function
 - inject dependency

"This won't scale because ..."

- file.read() / load all data

- streaming + yield

- one HTTP call per item
 - batching

- DB query in loop
 - N+1 problem

- heavy task in API
 - background worker / queue

- repeated LLM calls
 - cache (Redis)

"This blocks the event loop ..."

- requests / time.sleep in async
 - httpx.AsyncClient + await asyncio.sleep
- asyncio.run() in FastAPI
 - await directly

"I would add a timeout and a retry strategy..."

- requests.post(...)
 - timeout + retry + try/except + async client + idempotency key
- transient errors (502/503/timeout)
 - retry

"I would validate the input with Pydantic..."

- request.json() / dict from user
 - BaseModel DTO
- LLM JSON output
 - structured output + model_validate()
- tool calls
 - validate before execution

"This should be logged properly..."

- print()
 - logger
- except: pass
 - logger.exception()
- logs without context
 - trace_id
- logger.info(f"...")
 - lazy formatting

"I see a state management issue here..."

- self.chat_history in agent
 - stateless service + pass history in request

"I see a RAG issue here..."

- vector search without filter
 - tenant isolation

"I see an agent reliability issue here..."

- tool_calls[0]
 - check existence first

"I would move this to configuration..."

- hardcoded model/temp/max_tokens/retry
 - env/config file

"I would make this deterministic..."

- evals/benchmarks
 - fixed seed + temperature=0 + reproducible dataset